

# An interval for $S_{64}$ , the smallest Carmichael number with exactly 64 prime factors

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## Abstract

Let  $S_{64}$  denote the smallest Carmichael number with exactly 64 distinct prime factors (the  $k = 64$  entry of OEIS [A006931](#), currently unknown). We give an explicit two-sided bound. For the upper bound we exhibit a 148-digit Carmichael number  $N$  with exactly 64 prime factors, so  $S_{64} \leq N$ . For the lower bound we prove, by a finite exhaustion over a certified prime universe, that no 64-factor Carmichael number is smaller than  $10^{145}$ . Together,

$$10^{145} \leq S_{64} \leq N < 10^{148},$$

so  $S_{64}$  has 146, 147, or 148 decimal digits. All arithmetic is exact; the lower bound is a reproducible computation, verified here by three independent implementations, one of which takes its Korselt verdicts from a frozen reference oracle that shares no code with the search.

## 1 Statement

A *Carmichael number* is a composite  $n$  with  $b^n \equiv b \pmod{n}$  for all integers  $b$ . By Korselt's criterion,  $n$  is Carmichael iff it is squarefree and  $(p-1) \mid (n-1)$  for every prime  $p \mid n$ . Write  $\omega(n)$  for the number of distinct prime factors. The sequence  $S_k = \min\{n : n \text{ Carmichael}, \omega(n) = k\}$  is OEIS [A006931](#); values are known for  $3 \leq k \leq 63$ , and  $S_{64}$  is open. This note proves:

**Theorem 1** (Interval).  $10^{145} \leq S_{64} \leq N$ , where

$N = 866420555770562713589289732117698405485089408042974666135014101183659858491661736151818531159$

*is a 148-digit Carmichael number with exactly 64 prime factors. Consequently  $S_{64}$  has 146, 147, or 148 decimal digits.*

The two bounds are proved in §2 and §3. We use throughout the following elementary consequence of Korselt's criterion.

**Lemma 1** (Pairwise admissibility). *If  $p < q$  are distinct primes dividing a Carmichael number  $n$ , then  $p \nmid (q-1)$ . Moreover  $n$  is odd.*

*Proof.* Since  $q \mid n$  we have  $(q-1) \mid (n-1)$ , so  $n \equiv 1 \pmod{q-1}$  for every divisor  $d$  of  $q-1$ . If  $p \mid (q-1)$  then  $n \equiv 1 \pmod{p}$ ; but  $p \mid n$  gives  $n \equiv 0 \pmod{p}$ , forcing  $p \mid 1$ , a contradiction. For oddness: if  $n$  were even, an odd prime factor  $p$  would make the even number  $p-1$  divide the odd number  $n-1$  (as  $n$  is squarefree with an odd factor,  $n-1$  is even only if  $n$  is odd), which is impossible; concretely, no even number  $> 2$  is Carmichael.  $\square$

## 2 Upper bound

The number  $N$  of Theorem 1 is the product of the 64 primes

19, 29, 31, 37, 41, 43, 47, 53, 61, 67, 71, 73, 79, 89, 97, 101, 103, 109, 113, 127,  
131, 137, 139, 151, 157, 163, 167, 181, 193, 197, 199, 211, 239, 241, 251, 257, 271, 277, 281, 307,  
313, 331, 337, 353, 379, 397, 401, 421, 433, 449, 461, 463, 491, 541, 547, 577, 599, 631, 673, 811,  
829, 883, 1951, 3697.

These are 64 distinct primes; their product is  $N$  (re-derive  $N$  by multiplying the list rather than trusting the printed decimal), and one checks  $(p-1) \mid (N-1)$  for each. Hence  $N$  is a squarefree Carmichael number with  $\omega(N) = 64$ , and  $S_{64} \leq N$ . Since  $10^{147} \leq N < 10^{148}$ , the number  $N$  has 148 digits. The certificate is verified by `ref/ref_carmichael.py::verify_certificate`, an independent, dependency-free routine (independent Miller–Rabin primality, squarefreeness, and Korselt) sharing no code with the search that produced  $N$ .

**Remark 1.**  $N$  is an incumbent found by an exchange meet-in-the-middle search; it is not claimed to be minimal. It improves a previously reported 149-digit candidate and sits between Webster’s known neighbours  $N_{63}$  (145 digits) and the  $N_{65}$  candidate (151 digits), where a true  $S_{64}$  must lie.

## 3 Lower bound

Fix  $Y = 10^{145}$ . We show no 64-factor Carmichael number lies below  $Y$ .

### 3.1 A certified prime universe

**Lemma 2** (Cofactor lower bound). *Let  $n$  be a Carmichael number with  $\omega(n) = 64$ , and write its primes as  $p_1 < p_2 < \dots < p_{63} < q$ . Then*

$$D_n := \prod_{i=1}^{63} p_i \geq C, \quad C = 181214059509297180446495905961257618839971352500899762007368983685095561$$

*a 142-digit integer, independent of  $q$  and of any modulus.*

*Proof (finite computation).* The 63 cofactor primes are odd (Lemma 1) and pairwise admissible. Partition the argument on  $S$ , the set of cofactor primes that are  $\leq 97$ ; there are finitely many internally admissible such  $S$  (each pair in  $S$  satisfying Lemma 1). Given  $S$ , the remaining  $63 - |S|$  cofactor primes exceed 97 and are each individually admissible with every element of  $S$ ; replacing them by the  $63 - |S|$  *smallest* primes  $> 97$  that are individually admissible with  $S$  can only decrease the product, and *ignoring* the mutual admissibility constraints among those tail primes can only decrease it further. Thus for each admissible  $S$  we obtain a rigorous lower bound on  $\prod_i p_i$ ; taking the minimum over all 407760 admissible  $S$  gives the constant  $C$ . A finite prime table (odd primes  $\leq 10^4$ ) is used, and the computation checks that enough eligible tail primes are present for every  $S$ , so no smaller completion is truncated away.  $\square$

**Corollary 1** (Universe). *Every 64-factor Carmichael number  $n < Y$  has largest prime factor*

$$P^+(n) = q \leq \left\lfloor \frac{Y-1}{C} \right\rfloor = 5518.$$

*In particular every prime factor of such an  $n$  lies among the 727 odd primes  $\leq 5518$  (the largest being 5507).*

*Proof.*  $n = q D_n \geq q C$  by Lemma 2, and  $n < Y$  gives  $q \leq (Y-1)/C$ ; evaluate the floor.  $\square$

### 3.2 Exhaustion

**Theorem 2** (Lower bound). *There is no Carmichael number  $n < 10^{145}$  with  $\omega(n) = 64$ . Hence  $S_{64} \geq 10^{145}$ .*

*Proof (finite exhaustion).* By Corollary 1 every candidate is a product of 64 distinct primes drawn from the universe  $U$  of 727 odd primes  $\leq 5518$ , with product  $< Y$ . Enumerate all increasing 64-subsets of  $U$  with product  $< Y$ , pruning a partial choice only by two necessary conditions: (i) too few primes remain to reach 64; and (ii) the product of the current choice times the smallest admissible primes still available already reaches  $Y$  (an exact-integer  $R_{\min}$  test). Pairwise admissibility (Lemma 1) is enforced during descent—once  $p$  is chosen, every later prime  $q$  with  $p \mid (q - 1)$  is discarded—which removes no genuine Carmichael candidate. At each complete 64-product, test Korselt.

The enumeration terminates with exactly 127092 admissible complete products below  $Y$ , of which *none* is Carmichael. Therefore no 64-factor Carmichael number is smaller than  $10^{145}$ .  $\square$

### 3.3 Reproducibility

The lower bound was computed three independent ways, all agreeing:

1. A self-contained certificate (`s64_lower_bound_certificate.py`) that recomputes  $C$  and the universe, then runs *two* disjoint enumerators—a binary include/exclude walk using an incrementally maintained  $\text{lcm}(p_i - 1)$  for the Korselt test (327049 nodes), and an ordered-choice walk using direct per-prime divisibility, a separate prime generator, and no bitmasks (163525 nodes). Both report 127092 admissible products, 0 Carmichael, and an identical order-sensitive leaf digest `sha256 = e05f62598eea784b3f60f025c6ccbf30700115ecbef42702e24154dcece32321`.
2. An independent prime-bound tool (`carmichael_prime_bound.py`) that recomputes  $C$  and  $P^+ \leq 5518$  from first principles.
3. A separate enumerator that takes the Korselt verdict at every one of the 127092 leaves from the frozen oracle `ref/ref_carmichael.py` rather than from its own arithmetic, again finding 0 Carmichael. This enumerator was validated against the oracle’s brute-force `least_with_k` on small  $k$ :  $k = 3 \rightarrow 561$ ,  $k = 4 \rightarrow 41041$ ,  $k = 5 \rightarrow 825265$ .

## 4 The interval

Combining Theorem 2 (lower) with §2 (upper) proves Theorem 1:  $10^{145} \leq S_{64} \leq N < 10^{148}$ , so  $S_{64}$  has 146, 147, or 148 digits. This is a global statement about  $S_{64}$ ; it is *not* a determination of  $S_{64}$ , which would require excluding every 64-factor Carmichael number in the open interval  $(10^{145}, N)$ .

**Remark 2** (Next step). The same certified machinery run at  $Y = 10^{146}$  would either exhibit a smaller Carmichael number or prove  $S_{64} \geq 10^{146}$ , in the latter case narrowing the interval to 147–148 digits. The universe and the admissible-product count grow with  $Y$ ; whether the direct enumerator suffices at  $10^{146}$  or the exchange decomposition is required is the subject of ongoing work.

## Acknowledgements

The known values and neighbours for  $k = 3$ –63 (and the  $k = 65$  candidate) are from Jonathan Webster’s [small-carmichael-numbers](#) (Butler University for  $k = 36$ –63). The design of the lower-bound certificate benefited from consultation with a large-language reasoning model (OpenAI

GPT-6); the result reported here was independently reproduced by the three implementations of §3.3, including one gated by the frozen oracle. Existence of Carmichael numbers with every large factor count is due to Alford–Grantham–Hayman–Shallue and Larsen–Wright; the values above are new only as bounds on the *minimum*.